Yousef Kart

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Skills

Languages: C++17/20, Python, Java, TypeScript, SQL
Graphics: OpenGL, Qt 6, Three.js, ImGui, OpenCV
Tools: CMake, FastAPI, Git, Docker, Supabase, PostgreSQL

Experience

Red Piston April 2026 – Present
Full Stack Software Developer *Windsor, Ontario*

- Integrated Apple Health and Health Connect into MOVATI's mobile app, automating activity-based challenges and building a dashboard for health statistics
- Developed a multi-object 3D rendering feature for an augmented-reality platform used by 1,000+ users
- Built and shipped production web and mobile software across Next.js, React Native, Supabase, and Strapi
- Rebuilt a legacy Android application as a cross-platform React Native application for iOS, Android, and web, improving load performance and enabling real-time client monitoring

Scelta February 2025 – September 2025
Software Engineer *Windsor, Ontario*

- Developed a real-time 3D geospatial visualization pipeline for rendering development models on web-based maps
- Reduced REDMap's initial load time by 97% through database query optimization and API endpoint restructuring
- Led development of a shared UI component library used across multiple production applications, eliminating hundreds of duplicate files
- Designed a production-ready Next.js application architecture that reduced new-project build time by more than 50%

Freelance June 2021 – February 2025
3D Modeler *Remote*

- Delivered hundreds of optimized 3D assets for 40+ global clients, optimizing geometry, topology, and texture workflows for real-time performance
- Produced architectural visualizations for developments valued at $100M+, reducing asset turnaround times by 66%

Projects

Horizon 1 | *Python, FastAPI, PostgreSQL, LLMs, Finnhub, yfinance*

- Engineered an event-driven market intelligence pipeline that ingests financial news, classifies market events using LLMs, collects market data, and ranks potential trading opportunities
- Designed a scoring system combining event quality, relative volume, abnormal price movement, liquidity, market capitalization, and historical behaviour

Monte Carlo Simulation Engine | *C++20, OpenGL, GLFW, ImGui, CMake*

- Developed a stochastic simulation engine that uses Geometric Brownian Motion for Monte Carlo price modelling
- Built real-time OpenGL visualization and interactive ImGui controls for exploring simulated price paths

Black-Scholes Model Visualizer | *C++17, Qt 6, CMake*

- Implemented a Black-Scholes options-pricing engine with analytical computation of option Greeks
- Built interactive Qt visualizations for real-time viewing of option values and Greeks across market parameters

Flag Frenzy | *Lua, Roblox Studio, Blender*

- Built and launched a multiplayer capture-the-flag shooter with custom gameplay systems and optimized 3D content
- Scaled to 2.5M monthly visits before the project was acquired for $50K USD

Education

University of Windsor | *4x Dean's Honour Roll* September 2021 – June 2025
Bachelor of Computer Science (Honours) & Minor in Mathematics